

Discretionary Pensionary Benefits and Firm Performance: Evidence from India

Abstract

Discretionary pensionary expenses borne by firms form an important component of the total employee compensation. Such employee-benefits related expenses, however, go beyond and improve corporate image of a firm by signaling its financial soundness and improved ability to attract better talent. Corporate image, in turn, is associated with better firm performance. The expenses incurred on deferred compensation to employees also resemble profit sharing plans, and like them, are positively associated with firm performance. This paper tests association between discretionary pensionary expenses paid by firms on behalf of their employees and firms' financial performance. Results indicate that such expenses are indeed associated with firms' financial performance positively and significantly, whether an accounting or a market-expectation based measure is considered. The results obtained are robust to alternate measures of firms' financial performance or old-age social security expenses.

Keywords: Old-age social security, Provident Fund expenses, Financial Performance, Endogeneity, Corporate Image

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1 Introduction

Contribution towards old age social security of employees forms an integral part of the overall compensation structure within organizations. In India, apart from employees, employers are also mandated to contribute towards old age social security of their employees, which gets accumulated in the form of their provident fund (PF)¹. This state managed provident fund scheme has attributes of being both a defined contribution and a defined benefits scheme. Even as the contribution rate has been fixed at a certain percentage of employees' wages with a ceiling thereon, employers have a wide scope of restricting or widening the quantum of pensionary benefits to their employees, or even including or excluding them from receiving such benefits. It is, however, a fact that many firms pay towards the pensionary benefits on higher than mandated wages, even as they could have avoided such elevated payments. Moreover, such contributions do increase with time compared to a scenario when contributions are paid only at a mandatory wage ceiling². Such an action of beating rather than just meeting the mandatory requirement of pension funding presents a riddle. Are firms deriving some benefits if they undertake to commit substantial financial resources over prolonged periods? We argue that discretionary social security contribution expenses are positively associated with better firm performance. We also present the supporting empirical findings in this regard.

There is a vast literature in financial economics that links financial incentives to individual level performance outcomes (for a comprehensive review, please see Prendergast, 1999). As per agency theory, agents are averse to making efforts and need to be paid for motivating them to exert towards desired outcomes (Jensen & Meckling, 1976). Hence, the case for performance linked pay. But, agency theory does not explain the scenario when there is an overall, rather than task specific, increase in remuneration through deferred compensation element³, while payment of wages is not linked to some performance outcomes. Then there is the difficulty of linking the individual level performance outcomes to firm level outcomes, where the literature does not have a definitive say⁴. To some extent the discretionary pensionary expenses borne by firms in India resemble the popular deferred profit sharing schemes in the US. As per Kruse (1993), such profit sharing plans are a form of the defined contribution pension plans such as a 401(K) plan and enjoy similar taxation benefits, only difference is that these plans offer greater flexibility and discretion to employers as regards contribution. The major difference between a deferred profit sharing

¹ As provident funds can be termed as a form of defined contribution pension plan, we shall be generally terming them as a pension plan unless stated otherwise

² Such wage ceilings show inertia and increase only after a significant lag, e.g., wage ceiling was increased from INR 5000 in 1994 to INR 6500 in 2001, and then to INR 15000 in 2014.

³ Even in the extreme case when a firm were to pay out all the differential component of enhanced pensionary contribution directly to the employee in cash, the deferred component is still expected to weigh higher in view of highly attractive tax treatment accorded to such savings – it is tax deductible while investing, and does not attract any tax in accumulation and withdrawal stages.

⁴ Literature on High Performance Work Practices in strategic HRM says that an integrated set of well-managed HRM systems that are aligned with organizational objectives not only improve performance outcomes at individual level but also at the firm level (e.g. see Huselid, 1995; Huselid, Jackson & Schuler, 1997; Saridakis, Lai & Cooper, 2017). However, this literature requires fine-tuning of many HRM functions simultaneously and may not be apt for our analysis.

plan in the US and the discretionary contribution pension plan in India is that in the former, employees do not contribute to such schemes and periodic contributions by employers can vary as per the financial conditions⁵. In India, if a firm decides to pay pensionary contribution on enhanced reference wages, such contributions are sticky and do have a tendency to go up rather than down. Profit sharing plans have been found to be positively associated with firm performance (Doucouliagos, 1995; Kruse & Blasi, 1995; Prendergast, 1999; Arthur & Huntley, 2005). Drawing a parallel with the profit sharing plans, this paper explores the link between the discretionary pensionary expenses incurred by firms in India and their performance.

Motivation for the present study is also drawn by looking at another significant discretionary expenditure incurred by firms towards corporate social responsibility. One important facet is relationship of corporate reputation/image and corporate social responsibility (Du, Bhattacharya & Sen, 2010; Aguinas & Glavas, 2012). Brickley, Smith & Zimmerman (2002) and Jones (2005) argue that ethical behaviour or interactions with stakeholders positively impact the firm performance by first building a positive and enduring corporate image – an intangible that helps a firm reap the competitive advantage over ensuing time-periods. It has been established in the literature that corporate reputation is positively associated with better financial firm performance (Roberts & Dowling, 2002; Iwu-Egwuonwu & Chibuike, 2010). A firm making discretionary pensionary expenses is also expected to gain in reputation, which shows up in improved organization level performance.

This study utilizes a panel data of 16141 firm-years of the most liquid firms on the Bombay Stock Exchange (BSE)⁶ corresponding to the period 1990-2016. We use fixed effect OLS panel regression considering ROA, ROE, and Tobin's 'q' as response variables. Our study finds that the old age social security expenses borne by firms are indeed positively and significantly associated with firm performance measures mentioned above. We have considered both 'level' and 'growth' variables as regards the social security expenses. Our results are robust to different measures of performance, an alternate set of social security expense measures, and different sets of control variables used. To our knowledge, this is the first study that strives to empirically find an association between old age social security expenses and firm performance measures based on a panel data analysis.

Study has been arranged as followed - Section 2 presents a review of literature and also states the testable hypotheses, section 3 gives the data and variable definitions, Section 4 presents the methodology used in the study, section 5 presents the results and discussion thereon, Section 6 provides the results of various tests of robustness including the test of endogeneity, and Section 7 rounds up the study with conclusion.

⁵ For details, please refer to the link on IRS website <https://goo.gl/fDgDns> (accessed on 25/05/2017)

⁶ Belonging to Groups 'A' and 'B'

2 Literature Review and Hypothesis Development

2.1 Institutional Setup of Social Security Expenses in India

Many commentators consider old age social security schemes, such as pension plans, a tool to motivate and retain employees (McGill, 1996; Yousaf, Latif, Aslam & Siddiqui, 2014). There is another aspect to pension or provident fund provisioning. As early as 1913, DeRoode observed that provisioning of pension should not be considered only a humanitarian gesture shown by the management; rather it had strong economic repercussions because such firms attracted better employees and enjoyed the collective efficiency of the workforce by ensuring the permanence of employees. The same argument is extended as the reason why profit sharing plans do affect firm performance (Kruse, 1993; Kruse & Blasi, 1995). Such plans are expected to improve workers' productivity, while attracting better employees and retaining deserving ones.

Many pension plans are indeed designed in such a way that when an employee puts in less than a minimum number of years, his or her entitlement of receiving a pension is forfeited. The Employees' Pension Scheme (EPS), which is the most prevalent pension scheme in India and which the corporates mandatorily subscribe to, is such a scheme. If an otherwise eligible person puts in less than ten years of pensionable service, he or she is not entitled to receive a pension. In case a pension benefit plan is offered by a particular company as often happens in a country like the US, and if this plan stipulates a minimum service period before vesting pension, then such a scheme is expected to reduce employee turnover, which may, in turn, improve productivity because of learning and commitment possessed and shown by experienced employees. McGill makes a similar point that 'defined benefit pension plans'⁷ have often been used as 'golden handcuffs' so that an employee, on whom the firm has made significant investments in training and development over the years, should not leave the job. This also results in increased productivity as the cost of employee turnover is avoided (1996:340).

How lowering employee turnover and having experienced or trained employees may benefit a firm? Kleiner, Leonard, and Pilarsky (2002) suggest that the labour input that goes into the production function depends on the number of workers, number of hours put in by them, and the intensity of work rendered. A reduction in employee turnover mentioned above may favourably impact the production function by making the number of workers more stable and by maintaining or improving the work intensity by retaining experienced employees. There is, however, a caveat. A social security plan's impact on firm performance through improved productivity would fructify only if employees are well aware of such schemes and do care for them. As of now, we don't have much evidence to argue in favour of or against this proposition.

⁷ Such as EPS. A defined benefit plan computes the pension quantum on the basis of last drawn salary and length of service, as against a defined contribution plan, which offers pension by converting the accumulated savings in a designated account to an annuity.

An employee benefits scheme should prove more beneficial if it forms as integral part of the overall HRM systems and policies. This aspect was revealed in a qualitative research study carried out by Wood, Spinks, Leong, and Reeve (2010). They conducted 62 interviews with employers in the private sector after the passage of 'Pension Act 2008' in Great Britain. These employers were well spread out geographically and were of varying sizes; hence the sample could be considered representative, even though the sample size was small. The study found that such employers who felt it was employees' concern to save for their retirement were typically small in size, paid lower salaries, had a higher turnover of employees and often employed seasonal workers as compared to those employers who considered it their obligation to contribute towards their employees' pensions. The latter group of employers also considered pension contribution made by them as an integral part of their selection and retention policy and significant for employees' morale and productivity (pp. 2, 5, 15, 27, 42, and 45). In fact, many employers, who were already paying pension contribution (on behalf of their employees) at a higher rate than was the floor contribution rate stipulated in the Act, were reluctant to bring down their contribution rates as they feared that such an action would adversely impact employee morale and productivity.

2.2 Corporate Image, Corporate Social Responsibility, and Old-Age Social Security Contribution

Mandatory nature of subscription to the old age social security schemes⁸ in India means such firms that renege on the required deposition of old age social security contribution on behalf of their employees are open to administrative and legal actions from concerned government agencies. As per the annual report for 2015-16 of the Employees' Provident Fund Organization (EPFO) that administers the three integrated provident fund, pension, and insurance schemes, there were 31,445 cases pending for determination of dues from establishments that apparently failed to cover eligible employees and deposit contribution on their behalf on time; 28,546 prosecution cases were filed against various employers as regards non-deposition of such dues; and 65,970 recovery certificates were issued for collection of arrear demands⁹ from errant employers¹⁰, as on 31/03/2015.

A firm that is chary of its reputation will cautiously avoid being on the wrong side of a social security legislation, which has been duly mandated by the parliament. It can also be argued that such firms that don't want their reputations to be compromised would also be the firms that would strive to cultivate a good corporate image for themselves.

Apart from avoiding negative publicity, a firm that contributes at higher than mandated wage levels should gain in reputation. As per Brammer and Millington (2005), corporate reputation is a function of ability to attract and financial soundness. Fombrun and

⁸ Provident fund related expenses that provide coverage to an employee to three schemes simultaneously – an employees' provident fund scheme (EPF); a defined benefit pension scheme, viz., Employee' Pension Scheme (EPS); and an insurance scheme, viz., Employees' Deposit Linked Insurance Scheme (EDLI).

⁹ Recovery of such demands that remains pending for over one financial year is termed as arrear demand

¹⁰ Revenue certificates allow the authorities to take various pecuniary and legal actions, which may lead to seizure and sell of movable and/or immovable properties, attachment of bank accounts or even issuance of arrest warrants and confinement of responsible personnel to civil prison for commission of cognizable offence

Shanley (1990) said that managers seek to influence stakeholders' assessment by signaling their salient advantages. Discretionary pensionary contribution is a strong signal to attract prospective employees and is also a strong signal of financial soundness, and more importantly, stable future cash-flows.

While looking at discretionary pensionary expenditure on which not much of literature is present, it is pertinent to look at another significant discretionary expense undertaken by firms in the form of corporate social responsibility (CSR). Literature has established that the CSR activities undertaken by a firm help in firming up its corporate image, which helps in gaining financial returns (Orlitzky, Schmidt & Rynes, 2003; Du, Bhattacharya & Sen, 2010; Saeidi, Sofian, Saeidi, Saeidi & Saaeidi, 2015). Turban and Greening (1997) find that firms that rate high on corporate social performance (CSP) not only gain on corporate image but also are perceived favourably by prospective employees, which may end up giving such firms a competitive advantage. Bauman and Skitka (2012) mention that CSR activities help build up better overall impression about their employers in the eyes of employees. Turker (2009) found that organizational commitment of employees was higher if CSR activities were centered on employees, social/non-social stakeholders, or customers, but not on the government. Hence, if a CSR activity undertaken by a firm is perceived favourably by its employees, such a firm may reap financial dividends. Such an instance has been reported by Mishra and Suar (2010), who found that when managers were favourably disposed towards CSR activities, there was a positive association between such activities and firm's financial performance. One may argue that taking proper care of employee benefits should go hand in hand with CSR activities undertaken by a firm, and, hence, should have the same effect on firm's financial performance.

There is an established literature that finds a significant association between corporate reputation and firm's financial performance (see Dowling & Moran, 2012; Lee & Roh, 2012). Fombrun & Shanley (1990) point out that reputed firms can attract better applicants, get easier access to capital, can charge a premium for their products, and do attract investors. Firm's profitability is inevitably affected by these virtues that enhance the ease of doing business and offer a sustainable competitive advantage over the rivals. Roberts & Dowling (2002), in fact, find that corporate reputation provides returns not only in the immediate future but also on a sustained basis by giving a competitive advantage to a firm that possesses this intangible asset. The relationship between corporate image and firm's financial performance is, however, not unidirectional, e.g., Hammond & Slocum (1996) and Rose & Thomsen (2004) found that better financial performance leads to a better corporate image in the subsequent period. Earlier, McGuire, Schneeweis & Branch (1990) found a bidirectional relationship between reputation and financial performance. Nonetheless, causality is not an issue in our study; it would suffice if the corporate image, which in some ways is also a function of employee benefits offered by the firm, and firm performance, are associated in a positively significant way.

2.3 Hypothesis Development

In this study, we are interested in exploring the association between old-age social security expenses and firm performance. As per the discussion above, we expect such an

association to be significantly positive. Moreover, we believe that such association would exist not only for accounting measures of firm performance but also the market-based measure. The testable hypotheses stated in the alternate form are-

- (i) Both 'Growth' and 'level' variables related to old age social security expenses would be significantly and positively associated with accounting measures of profitability.*
- (ii) Both 'Growth' and 'level' variables related to old age social security expenses would be significantly and positively associated with market-based measure of profitability.*

3 Data and Variable Definition

3.1 Data

This study considers the financial data of firms belonging to BSE Group 'A' and 'B' as on 29/06/2016. The set comprised of 1545 most liquid firms in India having the largest market capitalization. Data has been collected from 'Prowess' database of 'Centre for Monitoring Indian Economy' (CMIE) for the period 1990-2016. Table-1 provides the sample description used in the analysis. The financial data is in the form of panel data having initial observations of 30741 firm-years. We have applied a few filters – we include a firm-year if only it is 12 months long and ends in March of a financial year as per the convention used in India. We then remove the firms having negative 'total assets' or 'book value of equity'. As the third level filter, we remove firms having negative or missing sales. Given the importance of 'profit after tax', 'provident funds expenses', and 'employee expenses' in the study, we also remove such firms that have missing values for these variables. We then remove financial services firms and utilities from our dataset at the fifth level of filtering. Finally, we remove all the firms that showed negative values for their 'age', which apparently is a wrong-coding-of-data problem. We are finally left with a sample size of 16141 firm-years for the period 1990-2016.

We have also obtained 'ownership' or 'holdings' related data of the firms mentioned above from Prowess. As this data is available only from the year 2001 onwards, we could take the pertinent data for the period 2001-2016 only. NIC codes of chosen firms have also been picked up from Prowess; however, for the purpose of removing firms belonging to financial services industries and utilities, we have used the coding nomenclature published by the Ministry of Corporate Affairs, Government of India¹¹.

Table-1 about here

¹¹ Available on http://www.mca.gov.in/MCA21/dca/efiling/NIC-2004_detail_19jan2009.pdf. Website accessed on 13/012/2016.

3.2 Variable Definitions

The complete list of dependent, independent, and control variables is given in Table-2. This table gives a description of variables used, their acronyms used in the study, their units, and their definition.

Table -2 about here

This study uses the performance measures 'Return on Assets' (ROA), Tobin's 'q' ratio, and 'Return on Equity' (ROE) as dependent variables. ROA and ROE are accounting measures, while Tobin's 'q' also takes into account the expectation of market as regards firm performance. All the three are widely established in the literature as firm performance measures; for instance, see Dess & Robinson (1984), Mehran (1995), Erhardt, Werbel & Shrader (2003), Dezso & Ross (2012), and Gul, Munir & Zhang (2016). As control variables, this study has again considered such variables that have been widely used in the literature. Accordingly, we use 'age' of a firm (Loderer & Waelchli (2010); Ekholm & Maury (2014)), it's 'size' given by $\log(\text{total assets})$ (Hermalin & Weisbach (1991); Dezso & Ross (2012)), 'leverage' given by 'total debt/total assets' (Mcguire, Sundgren & Schneeweis (1988); Aivazian, Ge & Qiu (2005)), 'sales growth' (Mcguire, Sundgren & Schneeweis (1988); Huselid, Jackson & Schuler (1997)), 'advertising intensity' given by 'advertising expenses/sales' (Mcguire, Sundgren & Schneeweis (1988); Hermalin & Weisbach (1991); Dezso & Ross (2012)), and 'research and development intensity' given by 'R&D expenses/sales' (Hermalin & Weisbach (1991); Mehran (1995); Peni (2014)).

Looking at the linkages between ownership structure and firm performance, this study also controls for ownership related variables in some of the models. There are studies that found association between ownership structure and accounting profitability measures, for example, Thomson & Pedersen (2000) found such evidence for large European companies, Xu & Wang (1999) for Chinese firms, Gedajlovic & Shapiro (2002) for Japanese firms, and Kumar (2004) found it in case of Indian firms. Evidence of a relationship between ownership and corporate valuation, often measured by Tobin's 'q', has also been there, for instance, see Morck, Shleifer & Vishny (1988), McConnell & Servaes (1990), Han & Su (1998). Accordingly, we have considered control variables 'insider' for promoters' shareholding, 'fii' for shareholding by foreign institutional investors, and 'retail' for holdings in the hands of the general public. We have also considered dummy variables depicting firms belonging to a business group, being a public sector undertaking, or having a foreign promoter and used them as controls to gather better insights on the relationship of ownership with firm performance, especially when provident fund expense related variables are expected to be influenced by ownership pattern as well.

Altman's 'z-score' is a well-regarded indicator of the risk of corporate bankruptcy and hence is a good measure of financial soundness of a firm ((Altman (1968, 2000), Altman, Iwanicz-Drozowska, Laitinen & Suvas (2016)). Gul, Munir & Zhang (2016) have used 'z-score' as an additional control variable for their analysis of firm performance of Malaysian firms. We have also used this derived variable as a control in one of our models.

Our independent variables of interest are old age social security expenses borne by firms in the form of provident fund contributions paid towards their employees. For the purpose of our analysis, we have considered both ‘level’ and ‘first level differenced’ variables. Natural logarithm of provident fund expenses (*‘log_provexp’*) is the key ‘level’ independent variables, while annual growth (log growth to be more precise) in the provident fund expenses, viz., *‘pfgrowth’* is the key ‘differenced’ independent variable. For robustness checks, we have also considered scaled provident fund expenses (*‘sc_provexp’*), given by ‘provident fund expense/total assets’ as the alternate ‘level’ independent variable, while scaled difference of provident fund expenses (*‘sc_diff_pf’*), has been taken as the alternate ‘differenced’ independent variable.

3.3 Correlations

Table-3 gives correlations among the variables used in the study and some additional variables of interest. We note that all the four provident fund expense related variables, whether level or differenced, are highly significantly (significance level 0.001) and positively correlated with all the performance measures, viz., *ROA*, *Tobin’s ‘q’*, and *ROE*. The only exception is *‘pfgrowth’* that shows correlation with *Tobin’s ‘q’* at 0.05 significance level when Pearson’s correlation is looked at; for Spearman correlation, even this correlation is significant at 0.001 level. We also observe that among the first set of control variables, viz., *‘log_ta’*, *‘sales growth’*, *‘adv_int’*, *‘rnd_int’*, *‘leverage’*, and *‘log_age’*, only *‘log_ta’* and *‘log_age’* are not significantly correlated to all the performance measures (both are, however, significantly correlated with *Tobin’s ‘q’*). Most of the ownership/holding related variables, viz., *‘insider’*, *‘fii’*, and *‘retail’* and also the dummy variables *‘bg’*, *‘psu’*, and *‘fgn’* are also significantly correlated with the performance measures, only exceptions are *‘bg’* with *ROA* and *ROE*, *‘psu’* with *‘q’* and *ROE*, and *‘fgn’* with *ROE* when looking at Pearson correlation coefficients, and *‘bg’* and *ROE* when referring to Spearman correlation coefficients. Expectedly, *‘z-score’* is highly significantly correlated with the three performance measures, whether looked at Pearson’s or Spearman’s correlation coefficients. It is also pertinent to note that the provident fund related variables are also significantly correlated with most of the control variables; hence we would need to check for multicollinearity issues while running our regression models.

Table-3 about here

3.4 Descriptive Statistics

Table-4 provides the descriptive statistics of all the variables used in the study. The table has been divided into three panels giving required statistics for dependent, independent, and control variables separately. We provide mean, first quartile, median, third quartile, standard deviation, minimum, maximum, and number of observations for each variable. All the variables and ratios including *ROA*, *ROE* and *Tobin’s ‘q’* have been winsorized at 1 and 99 percentiles on a yearly basis to check outliers in the data.

Table-4 about here

The mean ‘age’ of firms in the sample panel is 24.79 years, while the median is 24.02 years. Minimum age of a firm is 0 year, while the maximum age is 154 years¹². In the study, we have used natural log of age (*‘log_age’*) as a variable. **Leverage** in the firms across the years has mean and median values of 28.28% and 28.24% respectively. Minimum leverage is 0, i.e., we have debt free companies, while the maximum leverage is 86.48%. In fact, we have 1030 firm-years belonging to 229 unique firms when we have instances of zero leverage. The variable **scaled sales** (*‘sc_sales’*) suggest that mean and median sales are 0.94 times and 0.83 times the total assets respectively. Minimum ‘**sales to total assets value**’ is 0.01, while the maximum value is 3.96. Mean and median values of **total assets** (*‘ta’*) are INR 4359 million and INR 3984 million respectively. The smallest total asset base in the sample is INR 32.45 million, while the maximum is INR 798 billion. In our study, we have taken the natural logarithm of the total asset (*‘log_ta’*) as ‘**size**’ variable. Annual **sales growth** (*‘salesgrowth’*) in the sample shows a mean of 13.05% and median of 12.60%. The minimum yearly growth in sales is -160.7%, while the maximum is 299.99%. **Advertising intensity** (*‘adv_int’*) and **Research and Development intensity** (*‘rnd_int’*) show a mean value of 0.90% and 0.30% respectively, while even the upper quartile figures are a lesser 0.53% and 0.15% respectively. This means that both advertising and research and development expenses are highly skewed to the right implicating that a small number of firms are biggest spenders on these counts with respect to their sales. The mean score for **Altman’s z-score** (*‘z_score’*) in the panel dataset is 2.53, while the median is 2.26. On deeper analysis, it was found that 3557 firm-years corresponding to 734 unique firms have the z-score above 2.99, which means that these firms are in ‘non-bankrupt’ zone (Altman, 1968, p. 606). On the other hand, 5604 firm-years due to 995 firms have the z-scores between 2.99 and 1.81. As per the Altman’s nomenclature, these firms would fall in the ‘zone of ignorance’, which means they have chances of going bankrupt. This also means that an average Indian firm in our dataset, with a z-score of 2.53, doesn’t cross the upper floor set to be irrefutably financially sound, rather it falls in the gray zone being susceptible to corporate bankruptcy. In fact, there are 3975 firm-years belonging to 810 unique firms that have z-scores less than 1.81, the lower threshold set by Altman, meaning that these firms are more significantly vulnerable to corporate bankruptcy.

We have four provident fund (PF) expenses related variables – two of them are level variables, while the other two are first order differenced variables. The mean PF expense in our sample is INR 9.68 million, while the median is higher at INR 10.28 million meaning a negatively skewed distribution of these expenses across the firm-years. The minimum expense on PF by a firm is noted at INR 0.10 million, while the maximum is INR 2018.29 million. In the study, we have taken the natural logarithm of this variable (*‘log_provexp’*). **Growth in provident fund expenses** (*‘pfgrowth’*) is the key differenced variable we are considering. The sample shows a mean growth of 13.15% and median growth of 11.12% per annum in PF expenses. Minimum growth has been -207.80%, while the maximum rate obtained is 361.41%. The alternate level variable related to PF expenses is **scaled provident**

¹² Incidentally, it is ‘The Bombay Burmah Trading Company Limited’, which was founded in 1863. Company belongs to the Wadia group that also controls notable firms Bombay Dyeing and Britannia.

fund expenses (*'sc_provexp'*). It shows that the mean PF expenses are 0.00409 times the total assets, while the median is 0.00263 times the total assets. Minimum and maximum PF expenses in the dataset are 0.0001 and 0.042 times total assets respectively. The alternate first order difference variable used to check robustness is **scaled difference of provident fund expense** (*'sc_diff_pf'*). Mean of this variable is 0.00052, while the median is 0.00024. A minimum value is -0.01064, whereas the maximum is 0.01507. It is pertinent to mention here that the values of both *'sc_provexp'* and *'sc_diff_pf'* have been multiplied by 1000 to make them more legible and presentable in the table.

We are considering three performance-related measures, viz., return on assets ('ROA'), Tobin's 'q', and return on equity ('ROE'). The first two have been used in the main analysis, while ROE has been used as a robustness check. As per the table, we see that mean and median level ROE are 9.94% and 11.36% respectively, i.e., mean is lower than the median specifying a negatively skewed distribution. On the contrary, both ROA and Tobin's 'q' have means larger than their medians indicating positively skewed distributions. In fact, the positive skewness of Tobin's 'q' is more pronounced as the mean is almost equal to the upper quartile showing that there are a few ratio values that are decidedly high compared to the rest of the distribution. The minimum and maximum values of ROE are -280.4% and 85.26% respectively. We also notice that the standard deviation of ROE is more widely dispersed when compared to distributions of ROA and Tobin's 'q'. The coefficient of variation for ROE is 2.38, whereas it is only 1.31 for ROA and 1.76 for Tobin's 'q'. The mean values for ROA and Tobin's 'q' are 5.33% and 1.81 respectively, while the median values for the two are 4.52% and 1.05 respectively.

4 Methodology

For our study, we use panel data regression with industry and time fixed effects. We will be evaluating many models. For the fuller model, we will estimate the following regression:

$$Y_{it} = \beta_0 + \beta_1(pfgrowth)_{it} + \beta_2(log_provexp)_{it} + \beta_3(log_ta)_{it} + \beta_4(salesgrowth)_{it} + \beta_5(adv_int)_{it} + \beta_6(rnd_int)_{it} + \beta_7(leverage)_{it} + \beta_8(bg)_{it} + \beta_9(psu)_{it} + \beta_{10}(fgn)_{it} + \beta_{11}(insider)_{it} + \beta_{12}(fii)_{it} + \beta_{13}(retail)_{it} + \beta_{14}(z_score)_{it} + \gamma_2 I_2 + \gamma_3 I_3 + \dots + \gamma_{58} I_{58} + \delta_2 T_{1991} + \dots + \delta_{27} T_{2016} + u_{it}$$

Where,

– Y_{it} is the dependent variable (DV) where i = firm and t = time. DVs will be ROA, Tobin's 'q', or ROE.

– $(pfgrowth)_{it}, \dots, (z_score)_{it}$ represent independent variables (IV) and control variables and $\beta_1, \dots, \beta_{14}$ are the respective regression coefficients. In a few models, we will replace *'pfgrowth'* and/or *'log_provexp'* by *'sc_diff_pf'* and/or *'sc_provexp'*.

– u_{it} is the error term

– $I_2, \dots, I_{57}$ are the dummy variable representing two-digit NIC 2-digit industry codes. They are used in the model to obtain the industry fixed effects. There are 58 NIC 2-digit

codes for firms in our panel data; hence we have used 57 dummy variables. $\gamma_2, \dots, \gamma_{58}$ are the coefficients for these dummy variables.

– $T_{1991}, \dots, T_{2016}$ are time dummies used in the regression to obtain time fixed effects. As we are considering the period 1990 to 2016, we have used 26-time dummy variables in the study. $\delta_2, \dots, \delta_{27}$ are the coefficient for this time-dummy variable.

Fixed-effect model is used to control for any time-invariant omitted variables that might affect individual firms but no other firm in the panel taken in a cross-section. It, however, allows any such time-dependent effects that are same across all the firms in the sample (Hsiao, 2014). Considering that old age social security contribution related expenses are very much decided by firms individually, we consider fixed effect model more pertinent in this study. In robustness tests, we will also carry out Hausman test to check the suitability of fixed effect model vis-à-vis random effect model.

5 Results and Discussion

Tables-5.1 and 5.2 present the result of the fixed effect OLS regressions run on our panel data. Table 5.1 has 'Return on Assets' and Table 5.2 has Tobin's 'q' ratio as the dependent variable. In each of the tables, we have provided results of six regression models. Model -1 has the first order differenced variable '*pfgrowth*' and model-2 has level variable '*log_provexp*' as the independent variable. Both the models have '*log_ta*', '*salesgrowth*', '*adv_int*', '*rnd_int*', '*leverage*', and '*log_age*' as control variables. Models 3 to 6 have both the level and growth variables (i.e. both '*pfgrowth*' and '*log_provexp*') as independent variables as we want to check the individual effect of each variable in the presence of the other. Model-3 has same control variables as models 1 and 2. Model-4 expands the list of control variables by bringing in the dummy variables for firms belonging to a business group ('*bg*'), or those having government shareholding ('*psu*') or foreign shareholding ('*fgn*'). Model-5 further includes variables '*insider*', '*fii*', and '*retail*' signifying shareholding by promoters, foreign institutional investors, and retail investors respectively. Our objective is to observe how social security expenses are related to firm performance in companies that are owned differently. Model-6 is the full model, which has '*z_score*' as an additional variable, apart from all the variables present in model-5. In all the models, we have incorporated industry and year fixed effects.

Table-5.1 shows that both '*pfgrowth*' and '*log_provexp*' are significantly and positively associated with ROA, whether alone (in models 1 and 2) or together (in models 3, 4, and 5) at 0.001 significance level. The only exception is model-5, where '*pfgrowth*' is significant at a lower 0.01 level. It means that an increase in either growth rate of provident fund expenses or an absolute increase in the quantum of provident fund expenses is associated with an increase in the ROA. As per the results in model-5, an increase of one percentage point in '*pfgrowth*' will translate to 0.004 percentage point increase in ROA. Similarly, one unit increase in '*log_provexp*', which means 2.72¹³ units increase in provident fund expenses, will translate to 0.418 units increase in ROA. However, in model-6, when '*z_score*' is added as an additional variable, though '*pfgrowth*' remains highly significant (at

¹³ Value of the universal constant 'e' is 2.718281828459 up to 12 digits after decimal.

0.001 level), '*log_provexp*' becomes significant at 0.05 level only, but more importantly, it reverses sign. We also observe that advertising intensity ('*adv_int*'), which was expectedly positively and significantly related to ROA in models 1 to 5, becomes negatively significant in model 6 in the presence of '*z_score*'. Similarly, age variable ('*log_age*'), which was either insignificant (models 1 and 5) or expectedly negatively significant (models 2, 3, and 4), becomes positively significant in model-6. Another instance is of the intercept, which changes sign from being significantly positive in models 1 to 5 to significantly negative in model 6. In fact, the way '*z_score*' is calculated, it is expected to be very strongly associated with a performance measure (t-statistic 77.22 in this case) and, hence, may lead to somewhat counter-intuitive results. Another culprit may be a high correlation of '*z_score*' with '*log_provexp*' (0.13) as against with '*pfgrowth*' (0.05), though both are significant at 0.001 level. However, when we calculated variance inflation factor (VIF) for the model, the average VIF score was 2.22, with none of the variables showing a value greater than 10 (the highest being 4.61 for '*log_ta*'). This means that model is not undermined by the issue of multicollinearity. Despite this disruptive effect of '*z_score*' in model-6, '*pfgrowth*' remains positively significant (0.001 level) as mentioned above, while '*log_provexp*' also remains significant (0.05 level), though with reversed sign.

Table-5.1 about here

Looking at the control variables, we observe that the size ('*log_ta*') of a firm is positively and significantly associated with the accounting measure ROA, which means larger firms are associated with higher ROA. Similarly, sales growth ('*salesgrowth*') is positively significant (at 0.001 level) in all the six models, which also means that high-growth firms also return higher ROA. Advertising intensity ('*adv_int*') is highly and positively significant (again at 0.001 level) in all models except model-6 indicating that high advertising intensity goes with high ROA levels. As against advertising intensity, research and development intensity ('*rnd_int*') does not show an association with ROA. The analysis also shows that leverage is negatively significant (0.001 level) in all the models, which is again on expected lines. Higher leverage goes with lower ROA. Models 4, 5, and 6 show that firms that either belong to a business group ('*bg*'), are a public sector undertaking ('*psu*'), or have a foreign promoter, show lower ROAs. As against this, promoter shareholding ('*insider*') and foreign institutional investors' ('*fii*') interest in a firm are positively associated with ROA as per the models 5 ('*insider*' is positively significant in model 6 also). Shareholding by retail investors ('*retail*'), however, is not significantly related with ROA.

Summarily, the results obtained here support our first hypothesis that old-age social security expenses, whether 'level' or 'growth', are positively and significantly associated with ROA.

Table-5.2 presents the result of the panel regressions done using Tobin's 'q' as the dependent variable. The independent and control variables remain the same as they were in Table-5.1. As Tobin's 'q' considers expectations of the market, it is forward looking measure as against ROA or ROE that are accounting measures, and hence are based on past data. We notice that the level variable '*log_provexp*' remains positively significant in models 2 to 5 (at 0.001 level), but the growth variable '*pfgrowth*' ceases to be significant in any of the models.

It means market rewards those firms that spend higher quantum on provident fund and not necessarily those that show positive growth in such expenses. As per the results, in model-5, one unit increase in '*log_provexp*' (i.e., 2.72 unit increase in provident fund expenses) would result in 0.259 unit increase in the 'q' ratio. Model-6, however, shows that in presence '*z_score*' as an additional control variable, both '*pfgrowth*' and '*log_provexp*' cease to be significant.

Table-5.2 about here

As regards the control variables, we observe that sales growth ('*salesgrowth*') and advertising intensity ('*adv_int*') are significantly positive (at 0.001 level), while leverage is significantly negative in all the models as expected. Size ('*log_ta*') is positively significant (at 0.001 level) in models 1 and 6, but is negatively significant in model 4. Research and development intensity ('*rnd_int*') is notable as it is significantly positive (at 0.001 level) in all the six models. This variable was never significant (it was only significant in model-6, but negatively) when ROA was taken as the performance measure, but results in this table indicate that market rewards those firms that show higher R & D intensity. Age of a firm, which again was significantly negative in only three models when ROA was used, becomes significantly negative (at 0.001 level) in all the models except in model-6, when z-score is used as an additional control variable. As regards the holding related dummy variables, we saw that all the indicator variables for the business group ('*bg*'), PSU ('*psu*'), or foreign shareholding ('*fgn*') were negatively significant in models 4 and 5 when ROA was used as the DV. When a market-based measure Tobin's 'q' is used, indicator variables for business group and PSUs become insignificant, but one for foreign shareholding becomes significantly positive not only in models 4 and 5 but also in model 6. It means that market rewards firms with foreign shareholding and expects them to perform well in future. Moreover, firms belonging to a business group or having government shareholding are not penalised by the markets even as they are significantly negatively associated with ROA. As in case of ROA, Tobin's 'q' is also significantly and positively (at 0.001 level) associated with shareholdings of promoters ('*insider*') and foreign institutional investors ('*fii*') as per model-5 ('*fii*' shareholding is positively significant at 0.05 level in model-6 also). The variable representing retail shareholding ('*retail*') was insignificant when ROA was the DV, however, retail shareholding ('*retail*') is negatively significant in model-4 when Tobin's 'q' is used. Model-6 also shows that z-score is significantly positive (at 0.001 level).

Summarily, the results obtained here support our second hypothesis that old-age social security expense, given as level variable, is positively and significantly associated with Tobin's 'q'.

6 Tests for Robustness

6.1 Alternate Measure of Performance

We have used ROA and Tobin's 'q' as performance measures while running our regressions with the provident fund (PF) expense related independent variables and a

number of other control variables. To check whether PF expense related variables are significantly related to another performance measure, we ran the same six regression models using ROE as the firm performance measure. The results are presented in table-6.

Table-6 about here

We note that the growth of PF expense related variable (*'pfgrowth'*) is positively significant at 0.001 significant level in models 1, 3 and 4, and is positively significant at 0.05 significance level in model-6. The PF expense related level variable, viz., *'log_provexp'* is positively significant (at 0.001 significance level) in all the models except in model-6, which uses z-score as an additional control variable. Looking at the model-4, we observe that one percentage point increase in *'pfgrowth'* is associated with 0.015 percentage points increase in ROE. Similarly, one unit increase in *'log_provexp'*, i.e., 2.72 unit increase in PF expenditure is associated with 0.856 units increase in ROE. Looking at the relationship of performance measures with PF related expenses in six regression models run separately for ROA, Tobin's 'q', and ROE, one concludes that such old age security related expenses are indeed associated with performance measures, whether they are based on accounting measures or market expectation.

As regards the control variables, we see that significance level and signs of regression coefficients of variables mostly follow the results obtained in Table-5.1 for ROA. Size (*'log_ta'*), sales growth (*'salesgrowth'*), and advertising intensity are largely positively significant (exceptions are *'log_ta'* in models 2 and *'adv_exp'* in models 5 and 6). As in the case of ROA, R &D intensity is mostly insignificant. Leverage is highly and negatively significant (at 0.001 level) in all the models. Age (*'log_age'*) is also insignificant in all the models except in model-2, where it is negatively significant (at 0.05 level), and model-6, where it is positively significant at 0.05 level. This is at variance with the results obtained in regressions on ROA, where *'log_age'* was negatively significant in three models. Indicator variables showing whether a firm belongs to a business group (*'bg'*), is a PSU (*'psu'*), or has foreign shareholding (*'fgn'*), are all negatively significant as per models 4, 5 and 6 (at 0.001 level) except for insignificant *'bg'* in model-6. Ownership variables showing promoters' shareholding (*'insider'*) and FI's shareholding (*'fii'*) are positively and significantly associated with ROE. However, shareholding by retail investors (*'retail'*) is not significantly associated with ROE as per model 5. Altman's z-score is expectedly positive significant at 0.001 level of significance (model-6).

6.2 Alternate Level and Growth Variables Related to Old Age Social Security Expenses

We have established the association between variables related to provident fund expenses and firm performance measures ROA, Tobin's 'q', and ROE. We used *'log_provexp'* as the level variable and *'pfgrowth'* as the growth variable. A question may be raised whether the association would hold well even when a different set of growth and level variables are used as regards provident fund expenses. Accordingly, we chose two new independent variables. Scaled provident fund expense (*'sc_provexp'* given by provident fund expenses/total assets) is used as the level variable, while the Scaled difference of provident

fund expense (*'sc_diff_pf'* given by Δ (provident fund expense)/ (total assets)) has been taken as the growth variable. We have run the same set of fixed-effect panel data regressions with ROA and Tobin's 'q' respectively as response variables that we have used in earlier analyses. Results of the regressions are given in Table-7.1 (ROA as the dependent variable) and Table-7.2 (Tobin's 'q' as the dependent variable).

Table 7.1 shows that while the growth variable *'sc_diff_pf'* is positively significant (at 0.001 level) in all the six regression models, the level variable *'sc_provexp'* is also positively significant in models 2, 4 and 5 (at 0.001 level in models 2 and 5, and at 0.01 level in model 4). The level variable, however, is negatively significant (at 0.001 level) when we added z-score as an additional variable. We got similar results in earlier regressions run and reported in Table 5.1. These results validate our finding that PF expenses (both growth and level variables) are significantly associated with ROA.

Table -7.1 about here

As regards the control variables, we notice that size of a firm (given by *'log_ta'*) becomes positively significant (at 0.001 level) in all the models. Similar is the case with sales growth (*'salesgrowth'*) and advertising intensity (*'adv_int'*) – the only exception is *'adv_int'*, which becomes insignificant in model-6. R & D intensity is not significant in any of the models, except in model 6 (significant at 0.01 level). As in earlier models, leverage is negatively significant in all the six models (at 0.001 level). As regards the ownership/holding related variables, the dummy variables *'bg'*, *'psu'*, and *'fgn'* are negatively significant in models 4, 5, and 6 (exception is *'bg'* in model 6) in consonance with the results obtained in table-5.1. Similarly, promoters' shareholding (*'insider'*) and FII shareholding (*'fii'*) are positively significant (at 0.001 level), while retail shareholding is insignificant (model-5). As in earlier regressions, z-score is positively significant here also (at 0.001 level, in model 6).

Table-7.2, which uses Tobin's 'q' as the response variable, shows that both the growth and level P F expense related variables, viz., *'sc_diff_pf'* and *'sc_provexp'* are positively significant in all the six models, only exceptions being *'sc_provexp'* being insignificant in the presence of z-score in model 6. This is a departure from the results presented in Table-5.2 where the growth variable *'pfgrowth'* was insignificant in all the models. Looking at models 1 to 5, we also see that size (*'log_ta'*), sales growth (*'salesgrowth'*), advertising intensity (*'adv_int'*), and R&D intensity (*'rnd_int'*) are all positively significant (0.001 level). Similarly, leverage and age (*'log_age'*) are negatively significant in all the models 1 to 5. As noticed in table-5.2, dummy variables *'bg'* (indicating a business group) and *'psu'* (indicating government shareholding) are insignificant in both the models (*'bg'* is, however, positively significant in model-6 in the presence of additional variable *'z-score'*), while *'fgn'*, standing as indicator variables for foreign shareholding is significantly positive in all the models whenever included (i.e. models 4,5, and 6). Ownership related variables showing promoters' shareholding (*'insider'*) and FII's shareholding (*'fii'*) are significantly positive, while retail investors' shareholding (*'retail'*) is negatively significant (all at 0.001 level) as per model-5 as noticed in table-5.2. Altman's z-score (*'z_score'*) is significantly positive (at 0.001) as per model-6. Summarily, the results here agree with those given in table-5.2, the only difference

is that the association between the performance measure and PF expense related variables is more pronounced in this case with a new pair of growth and level variables.

Table-7.2 about here

6.3 Tests for heteroscedasticity and serial correlations

We are using a large sample of over 16000 firm-years during a period of 27 years (1990-2016). In such a panel, issues of heteroskedasticity are likely to crop up. To check for group-wise heteroskedasticity, we ran modified Wald test on models 5 and 6 using both ROA and Tobin's 'q' as response variables separately. In all the circumstances, the test results were highly significant (p-value 0.0000; the null being the data is homoscedastic), showing that the sample data has heteroskedasticity. We also tested for serial correlations in the data. For this purpose, we conducted the Wooldridge test for autocorrelation in panel data, where the null is that there is no first-order autocorrelation. We again ran the test for models 5 and 6 using ROA and Tobin's 'q' separately. All the test results showed that the data showed serial correlation (p-value 0.0000). Accordingly, we tested the six regression models again with Huber-White sandwich variance-covariance estimator to use robust standard errors and taking account of serial correlations. As before, ROA and Tobin's 'q' have been used as dependent variables. The test results are shown in Tables 8.1 and 8.2.

Table-8.1 shows that even as t-statistics have come down compared to those mentioned in table-5.1, the results are largely in conformation with earlier results. Both the growth and level variables ('*pfgrowth*' and '*log_provexp*' respectively) are positively significant in all the six models with the only exception of '*log_provexp*' being insignificant in mode-6. The growth variable is significant at either 0.001 or 0.01 significance level. The level variable is significant at 0.05 level in models 1 to 5. We conclude that the relationship between P F expense related variables and ROA remains significant and positive even when robust standard errors are used.

Table-8.1 about here

As regards the control variables, we notice that size ('*log_ta*') is significant in models 1 (at 0.001 level) and 4 (at 0.10 level). In other models, it is not significant. As in earlier models, sales growth ('*salesgrowth*') is positively and leverage is negatively significant in all the six models (0.001 level). Advertising intensity ('*adv_int*') is no more significant, except in model-1, where it is significant at 0.10 level. R&D intensity ('*rnd_int*'), as before, is not significant in any of the models. Age of firm ('*log_age*') is insignificant, except in model-2, where it is significant at 0.10 level. Indicator variables showing whether a firm belongs to a business group ('*bg*') or has government shareholding ('*psu*') are negatively significant at 0.001 level as before (in models 4 and 5). The dummy variable showing whether a firm has foreign shareholding is also negatively significant at 0.01 level in model 4, and at 0.10 level in model 5. Variables '*insider*' and '*fii*' are positively significant as before (at 0.01 and 0.001 level respectively) in both models 4, and 5 and so is z-score in model-6 (at 0.001 level). Even as a few control variables have become insignificant, PF expense related variables are still significant in most of the models underlying their strong association with ROA.

Table-8.2 shows that as before in table 5.2, the level PF expense related variable, i.e., '*log_provexp*' is positively significant in all the models at either 0.01 or 0.001 level of significance. It is insignificant only in model-6, where z-score is inducted as an additional control variable. The growth variable is positively significant in model 1 (at 0.10 level), which was not the case earlier as reported in table-5.2. In control variables, sales growth ('*salesgrowth*') and leverage are highly significant in all the models (at 0.001 level; the only exception is sales growth that is significant at 0.05 level in model-6). Advertising intensity ('*adv_int*') is positively significant at 0.10 level in models 1, 2, 3, 4, and 6, which is a departure from the earlier analysis presented in Table -5.2, which showed that it was highly significant at 0.001 level. R&D intensity ('*rnd_int*') was also significant when market-based performance measure Tobin's 'q' was used (see Table -5.2), but here it is insignificant in all the models. The coefficient of age variable ('*log_Age*') is consistently negative and is significant in models 2, 3, and 4 (at 0.05 level). This shows that market expects older firms to perform poorly. Holding related dummy variables are mostly insignificant unlike in the case of ROA. Only exceptions are '*bg*' being positively significant in model-6 (at 0.05 level) and '*fgn*' being significant in model 4, 5, and 6 (0.05 level in models 4 and 6, and 0.10 level in model 5). As in the case of ROA, '*insider*' and '*fii*' positively significant at 0.01 and 0.001 level respectively in model-5; while '*retails*' is negatively significant at 0.10 level in model-5. Altman z-score is positively significant (at 0.001 level).

Table-8.2 about here

6.4 Fixed versus Random Effect

We have used fixed effect model in our panel dataset as we wanted to observe within group variations rather than between groups. We have, however, carried out Hausman's test to check the suitability of using the fixed effect model vis-à-vis random effect model. The null being tested is that there is no difference between estimators obtained from fixed or random effects. The test result has been obtained using three regression models-3, 5, and 6 for each of the response variables – ROA, Tobin's 'q', and ROE. Results have been given in Table-9, which gives the threshold probability values for Chi-square statistic obtained in the test. We notice that barring one regression (ROA for model-3), all the results are significant at 0.0000 level. We, therefore, reject the null hypothesis and conclude that the fixed effect model is preferable to use in our panel data.

Table-9 about here

6.5 Tests of Endogeneity

As happens in such analyses, the relationship among variables being explored in this study may suffer from endogeneity problem. Performance measures, i.e., ROA, Tobin's 'q', and ROE may influence PF expense related variables – both level and growth. It can well be argued that financial sound firms would be in a better position to take care of their employees and, being an important measure of overall social security and employee welfare, provident funds may also show consonant increments. Accordingly, we ran two-stage least square (2-SLS) regressions using instrumental variables to take care of the potential problem of endogeneity. We have used total employee compensation net of provident fund expenses

(*netempexp*) as an instrumental variable for the level PF expense variable '*log_provexp*', and growth in employee expenses (*empexpgrowth*) as an instrumental variable for the growth-related PF expense variable '*pfgrowth*'. In the first stage we first predicted the values of our independent variables (PF expense related variables) using instrumental variables, and then, in the second stage, these predicted values of the independent variables were regressed with the three performance measures as dependent variables. Table-10.1 reports the result obtained for running model-5 in two stages for all the performance measures ROA, Tobin's 'q', and ROE.

Table 10.1 about here

We note that both '*pfgrowth*' and '*log_provexp*' are positively significant (at 0.001 significance level) in all the three results presented here. The control variables also show similar significance levels as observed in earlier regressions. We, therefore, conclude that the results obtained from Instrumental Variable-two stage least square (IV-2SLS) are consistent with our previous findings.

6.5.1 Testing for endogeneity of variables

We estimate Durbin's and Wu-Hausman's test statistic to check whether variables are exogenous. These statistics were computed for model-5, for all the three performance measures ROA, Tobin's 'q', and ROE. The null hypothesis to be tested was that variables are exogenous. Results of the tests are given in table-10.2. We see that all the results are highly significant (probability value 0.0000). The null is, therefore, rejected and we conclude that the variables (i.e., '*pfgrowth*' and '*log_provexp*') are indeed endogenous. Hence, our findings presented in table-10.1 get necessitated and further validated.

Table 10.2 about here

6.5.2 Test for strength of instrumental variables used

While choosing instrumental variables, we need to ensure that variables selected are not weak. As per Wooldridge, a good exogenous instrument should be correlated with the endogenous variable, but not with the error terms (2013: 514). We, therefore, estimate the extent of correlation of our endogenous variables and additional instruments. We evaluate F-statistic showing the joint significance of regression coefficients of instrumental variables in the first stage regressions, done using the endogenous variables as response variables. A significant statistic shows that instruments used are strong. Table-10.3 shows the F-test statistic for model-5 regressed with ROA and Tobin's 'q' (ROE and ROA give the same statistic). All the statistics are highly significant (probability value 0.0000) underlying the fact that instruments used are not weak.

Table 10.3 about here

As we have the same number of endogenous variables as instrumental variables, we do not need to run tests for over-identification of instrumental variables.

7 Conclusion

The study confirms Provident Fund expense related variables are significantly and positively associated with firm performance variables, whether they are accounting measures ROA or ROE, or a measure informed by market expectations, viz., Tobin's 'q'. We noticed that whereas the level variables, whether natural logarithm of provident fund expenses or scaled provident fund expenses, were always positively significant at high significance level, the growth variable, given by the yearly growth in provident fund expenses, was insignificant only in case of Tobin's 'q'. However, the alternate growth variable used by us, viz., scaled difference in the provident fund expenses, was positively significant for Tobin's 'q' as well. This result may mean several things - a firm that designs attractive provident fund plans for its employees, also becomes alive to other welfare measures of its employees and creates a more conducive workplace environment resulting in better productivity and firm performance. One may also argue that provident fund expenses grow in the proportion of overall compensation and hence act as a potent motivator for employees, which again gets reflected in enhanced firm performance. These arguments also present the limitations of the study - we are not essentially aware how provident fund expenses are actually associated with performance outcomes. Deciphering this linkage should be an attractive future course of work. HR managers, however, should consider the results of the study in greater detail and should design more beneficial social security plans, not to talk of restricting the payment outgo on even the government mandated contribution, as it might reflect in their firms' financial performance.

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Table-1 (Sample selection)

Sampling description

	Number of firm years
Total sample size for the period 1990-2016	30741
Less: Financial years having length different from 12 months and/or not ending in March	3760
Less: Firms with negative or missing 'total assets' or 'book value of equity'	924
Less: Firms with negative or missing sales	1828
Less: Firms with missing Profit After Tax (PAT), Employee Expenses, or Provident Funds Expenses	6837
Less: Financial companies and utilities in the datasets	1242
Less: Firms have negative value for 'age'	9
Final Sample Size	16141

Table-2 (Variable Definitions)

This table gives the definition of dependent, independent, and control variables used in the analysis.

Variable	Acronym used	Unit	Definition
Natural logarithm of Age	<i>log_age</i>	Age in number of years	current financial year - incorporation year + 1
Leverage	<i>leverage</i>	Percentage	(total debt/total assets)*100
Scaled Sales	<i>sc_sales</i>	Ratio	sales/total Assets
Natural logarithm of total assets	<i>log_ta</i>	Total assets in INR Million	ln(total assets)
Growth in sales	<i>salesgrowth</i>	Percentage	ln(sales(t)/sales(t-1))*100
Advertisement Intensity	<i>adv_int</i>	Percentage	(advertisement expenses/total sales)*100
Research and Development Intensity	<i>rnd_int</i>	Percentage	(research and development expenses/total sales)*100
Altman's z-score	<i>z_score</i>	Ratio	1.2*working capital/total assets + 1.4*reserves/total assets + 3.3*earnings before interest and taxes /total assets + 0.6*market capitalization/book value of equity + total sales/total assets
Natural logarithm of provident funds (PF) expenses	<i>log_provexp</i>	PF expenses in INR Million	ln(provident funds expenses)
Growth in provident funds expenses	<i>pfgrowth</i>	Percentage	ln(provident funds expense(t)/provident funds expense(t-1))*100
Scaled provident funds expenses	<i>sc_provexp</i>	Ratio	provident funds expenses/total assets
Scaled difference in provident fund expenses	<i>sc_diff_pf</i>	Ratio	(provident funds expenses(t) –provident funds expenses(t-1))/total assets(t)
Insider shareholding	<i>insider</i>	Percentage	Shows the proportion of equity shares (in terms of percent) that are held by the promoters of the company at the end of a financial year. It includes both Indian and foreign promoters.
FII shareholding	<i>fii</i>	Percentage	Shows the proportion of equity shares (in terms of percent) of the company that are held by foreign institutional investors at the end of a financial year.
Retail shareholding	<i>retail</i>	Percentage	Shows the proportion of equity shares (in term of percent) of the company that are held by individual non-promoter investors at the end of a financial year.
Dummy variable for foreign ownership	<i>fgn</i>	0 or 1	1 if foreign ownership is not 0
Dummy variable for government ownership	<i>psu</i>	0 or 1	1 if government ownership is not 0
Dummy variable for business group	<i>bg</i>	0 or 1	1 if firm belongs to a business group
NIC 2-digit code	<i>nic2d</i>	2 digits	First 2 digits of NIC code showing industry which a firm belongs to
Return on Equity	<i>roe</i>	Percentage	(profit after tax/book value of equity)*100
Return on Assets	<i>roa</i>	Percentage	(Profit after tax/total assets)*100
Tobin's q ratio	<i>q</i>	Ratio	(market capitalization + total debt)/(book value of equity + total debt)

Table-3 (Correlations)

This table gives both Pearson's and Spearman's correlation among the variables involved. The upper triangle provides the values of correlation coefficients obtained using Pearson method, while the lower triangle gives the values of Spearman's rank correlation. Stars provide the significance levels as per the notation mentioned below the table. The sample period is 1990 to 2016, and the total sample size is 16141 firm years.

	pfgrowth	log_prove	sc_prove	sc_diff_pf	log_ta	salesgrowth	adv_int	rnd_int	leverage	log_age	bg	psu	fgn	insider	fii	retail	z_score	empexpgr	netemp	sales	roa	q	roe
pfgrowth	1.00***	0.05***	0.05***	0.60***	-0.01	0.27***	0.02	0	0.02	-0.13***	-0.05***	0	-0.01	0.03*	0.04***	-0.07***	0.05***	0.45***	0	0	0.10***	0.03*	0.09***
log_prove	0.03***	1.00***	0.34***	0.13***	0.75***	-0.05***	0.07***	0.18***	-0.08***	0.44***	0.29***	0.25***	0.05***	0	0.34***	-0.35***	0.13***	-0.06***	0.51***	0.44***	0.12***	0.17***	0.06***
sc_prove	0.06***	0.44***	1.00***	0.40***	-0.21***	-0.01	0.09***	0.09***	-0.18***	0.16***	0.08***	0	0.08***	0	-0.11***	0.10***	0.23***	-0.05***	0.06***	-0.07***	0.14***	0.11***	0.07***
sc_diff_pf	0.84***	0.18***	0.40***	1.00***	-0.09***	0.21***	0.03*	0.02	-0.06***	-0.09***	-0.04***	-0.01	0.01	0.02	0.02	-0.03	0.14***	0.31***	0.02	-0.02	0.14***	0.08***	0.11***
log_ta	-0.01	0.74***	-0.22***	-0.09***	1.00***	-0.05***	0.01	0.10***	0.04***	0.32***	0.24***	0.28***	-0.01	-0.02	0.49***	-0.48***	-0.03*	-0.04***	0.51***	0.54***	0.01	0.11***	-0.02
salesgrowth	0.27***	-0.08***	-0.04***	0.23***	-0.05***	1.00***	-0.01	-0.03*	0.01	-0.14***	-0.03*	-0.02	-0.01	0.01	0.06***	-0.08***	0.18***	0.54***	-0.02	-0.01	0.26***	0.08***	0.28***
adv_int	0.03***	0.03***	0.07***	0.06***	-0.01	0.03**	1.00***	0.01	-0.13***	0.02	0.04***	-0.07***	0.04***	0.06***	0.15***	-0.09***	0.21***	0.03*	-0.01	-0.02	0.09***	0.21***	0.04***
rnd_int	-0.05***	0.41***	0.32***	0.08***	0.21***	-0.05***	0.02**	1.00***	-0.10***	0.06***	0.04***	0.01	-0.01	-0.02	0.11***	-0.02	0.10***	-0.01	0.11***	0.03**	0.07***	0.16***	0.03*
leverage	0.02**	-0.07***	-0.17***	-0.05***	0.05***	0.02**	-0.12***	-0.09***	1.00***	-0.07***	0.02	-0.11***	-0.14***	-0.05***	-0.13***	0.09***	-0.47***	0.03**	-0.13***	-0.07***	-0.44***	-0.23***	-0.28***
log_age	-0.15***	0.46***	0.24***	-0.05***	0.33***	-0.16***	0	0.33***	-0.06***	1.00***	0.21***	0.10***	0.04***	-0.03**	0.02	-0.03*	0.01	-0.20***	0.17***	0.15***	0.02	0.02	0.01
bg	-0.06***	0.30***	0.11***	-0.02**	0.25***	-0.05***	-0.03***	0.16***	0.02**	0.23***	1.00***	-0.21***	-0.22***	-0.07***	0.10***	-0.03	-0.04***	-0.06***	0.06***	0.03**	-0.01	0.03*	0.01
psu	-0.02	0.23***	0.01	-0.02**	0.24***	-0.03**	-0.05***	0.07***	-0.12***	0.11***	-0.21***	1.00***	-0.04***	0.19***	0.02	-0.23***	0.06***	-0.03**	0.28***	0.35***	0.04***	0.03	0.02
fgn	-0.01	0.05***	0.09***	0.02*	-0.01	-0.02*	-0.02*	0.04***	-0.14***	0.04***	-0.22***	-0.04***	1.00***	0.10***	-0.01	-0.05***	0.10***	-0.02	0.01	0	0.05***	0.08***	0.01
insider	0.04***	-0.02	0.02**	0.04***	-0.02**	0	0.04***	-0.05***	-0.06***	-0.03***	-0.08***	0.16***	0.11***	1.00***	-0.24***	-0.46***	0.13***	0.03	-0.03	0	0.08***	0.08***	0.06***
fii	0.06***	0.44***	-0.13***	0.01	0.60***	0.07***	0.12***	0.11***	-0.15***	0.05***	0.13***	0.05***	0.02*	-0.18***	1.00***	-0.38***	0.20***	0.09***	0.30***	0.29***	0.17***	0.21***	0.09***
retail	-0.08***	-0.34***	0.14***	-0.02*	-0.49***	-0.08***	-0.07***	0.02**	0.11***	-0.02	-0.01	-0.25***	-0.05***	-0.44***	-0.43***	1.00***	-0.18***	-0.10***	-0.25***	-0.26***	-0.16***	-0.18***	-0.10***
z_score	0.09***	0.10***	0.28***	0.20***	-0.11***	0.23***	0.15***	0.16***	-0.51***	0.02	-0.06***	0.07***	0.10***	0.14***	0.15***	-0.17***	1.00***	0.14***	0.13***	0.13***	0.69***	0.63***	0.48***
empexpgrowth	0.46***	-0.07***	-0.07***	0.38***	-0.03***	0.52***	0.06***	-0.06***	0.03***	-0.22***	-0.08***	-0.05***	-0.02*	0.02*	0.11***	-0.11***	0.19***	1.00***	-0.02	-0.02	0.21***	0.08***	0.22***
netempexp	-0.01	0.93***	0.27***	0.08***	0.81***	-0.07***	0.04***	0.36***	-0.09***	0.42***	0.26***	0.23***	0.05***	0.01	0.51***	-0.39***	0.10***	-0.03***	1.00***	0.79***	0.12***	0.13***	0.05***
sales	0.01	0.78***	-0.03***	0	0.87***	0.03***	-0.03***	0.28***	0.04***	0.36***	0.22***	0.25***	0	0.02*	0.50***	-0.44***	0.14***	0.02*	0.83***	1.00***	0.08***	0.12***	0.05***
roa	0.14***	0.12***	0.17***	0.21***	0	0.28***	0.11***	0.15***	-0.44***	0.01	-0.02*	0.04***	0.06***	0.08***	0.20***	-0.15***	0.73***	0.27***	0.13***	0.14***	1.00***	0.28***	0.75***
q	0.11***	0.32***	0.13***	0.16***	0.25***	0.23***	0.15***	0.19***	-0.22***	0.10***	0.06***	0.07***	0.10***	0.11***	0.42***	-0.37***	0.57***	0.24***	0.35***	0.32***	0.49***	1.00***	0.13***
roe	0.16***	0.10***	0.14***	0.21***	0.01	0.36***	0.08***	0.15***	-0.22***	0.01	-0.01	0.05***	0.02*	0.08***	0.16***	-0.14***	0.62***	0.31***	0.11***	0.18***	0.90***	0.51***	1.00***

p < .001, "***", p < .01, "**", p < .05, "*", p >= .05, " "

Upper triangular matrix shows Pearson's correlation coefficients, while the lower triangle gives Spearman's.

Table-4 (Descriptive Statistics)

This table provides descriptive statistics of variables of interest. 'Age' is in number of years. The level variables, viz., 'age', 'sales', 'ta', and 'provexp' are in Indian Rupee Million. 'Leverage', 'salesgrowth', 'pfgrowth', 'adv_int', 'rnd_int', 'roe', and 'roa' are in percentage. 'sc_sales', 'z_score', 'sc_provexp', 'sc_diff_pf', and 'q' are ratios. 'sc_procexp' and 'sc_diff_pf' have been multiplied by 1000 for better readability.

Variable	mean	Lower quartile	median	Upper quartile	Standard dev	min	max	N
<i>Panel-A (Firm Characteristics)</i>								
<i>log_age</i>	3.25	2.83	3.22	3.69	0.65	0.00	5.04	16141
<i>leverage</i>	28.28	12.77	28.24	41.94	18.97	0.00	86.48	16141
<i>sc_sales</i>	0.94	0.52	0.83	1.21	0.61	0.01	3.96	16141
<i>log_ta</i>	8.38	7.25	8.29	9.42	1.68	3.48	13.59	16141
<i>salesgrowth</i>	13.05	1.24	12.60	24.74	29.15	-160.17	219.99	14466
<i>adv_int</i>	0.90	0.00	0.03	0.53	2.20	0.00	18.35	16139
<i>rnd_int</i>	0.35	0.00	0.00	0.15	1.11	0.00	10.33	16137
<i>z_score</i>	2.53	1.67	2.26	3.08	1.40	-9.35	24.50	13136
<i>Panel-B (Provident Fund related variables)</i>								
<i>log_provexp</i>	2.27	1.06	2.33	3.54	1.86	-2.30	7.61	16141
<i>pfgrowth</i>	13.15	0.00	11.12	27.51	40.97	-207.80	361.41	14462
<i>sc_provexp</i>	4.09	1.09	2.63	5.43	4.52	0.01	42.19	16141
<i>sc_diff_pf</i>	0.52	0.00	0.24	0.84	1.75	-10.64	15.07	14466
<i>Panel -C (Firm Performance related variables)</i>								
<i>roe</i>	9.94	4.03	11.36	19.80	23.71	-280.40	85.26	16141
<i>roa</i>	5.33	1.57	4.52	8.66	6.98	-25.57	34.88	16141
<i>q</i>	1.81	0.77	1.05	1.83	3.19	0.03	127.06	13983

Table-5.1 (RoA as Firm Performance measure)

This table provides the result of regression on the panel data considering 'RoA' as the dependent variable. Model (1) uses 'pfgrowth' as the independent variable, while considering 'log_ta', 'salesgrowth', 'adv_int', 'rnd_int', 'leverage', and 'log_age' as control variables. Model (2) uses the level 'log_provexp' as the independent variable, while using the same control variables as in model (1). Model (3) uses both 'pfgrowth' and 'log_provexp' as independent variables and has the same controls as in previous models. Model (4) expands model (3) by including ownership dummy variables 'bg', 'psu', and 'fgn'. Model (5) further adds 'insider', 'fii', and 'retail' holdings as control variables. Finally, the model (6) gives the full model by also incorporating z_score as a control variable. Regression coefficients have been given along with significance levels; and t-statistics have been reported in the parenthesis below the coefficients.

	(1)	(2)	(3)	(4)	(5)	(6)
<i>pfgrowth</i>	0.007*** (6.27)		0.006*** (4.98)	0.005*** (4.58)	0.004** (2.95)	0.004*** (3.75)
<i>log_provexp</i>		0.386*** (8.43)	0.345*** (7.43)	0.442*** (9.43)	0.418*** (8.36)	-0.085* (-2.10)
<i>log_ta</i>	0.444*** (13.09)	0.125* (2.44)	0.156*** (3.04)	0.260*** (4.98)	0.126* (2.02)	0.512*** (10.01)
<i>salesgrowth</i>	0.053*** (30.60)	0.055*** (33.41)	0.053*** (30.87)	0.052*** (30.70)	0.055*** (30.10)	0.027*** (17.26)
<i>adv_int</i>	0.214*** (9.21)	0.199*** (8.51)	0.198*** (8.48)	0.190*** (8.10)	0.145*** (5.83)	-0.044* (-2.20)
<i>rnd_int</i>	-0.029 (-0.63)	-0.068 (-1.45)	-0.064 (-1.38)	-0.066 (-1.42)	-0.082 (-1.76)	-0.119** (-3.16)
<i>leverage</i>	-0.175*** (-62.86)	-0.172*** (-61.12)	-0.173*** (-61.31)	-0.178*** (-62.48)	-0.180*** (-59.81)	-0.091*** (-33.48)
<i>log_age</i>	-0.14700 (-1.71)	-0.419*** (-4.67)	-0.357*** (-3.94)	-0.284** (-3.13)	0.050 (0.52)	0.201** (2.58)
<i>bg</i>				-0.863*** (-7.59)	-0.884*** (-7.45)	-0.123 (-1.29)
<i>psu</i>				-3.404*** (-10.93)	-4.338*** (-12.86)	-3.168*** (-11.59)
<i>fgn</i>				-1.262*** (-4.93)	-1.326*** (-4.97)	-1.114*** (-5.26)
<i>insider</i>					0.039*** (9.65)	0.010*** (3.13)
<i>fii</i>					0.079*** (9.62)	-0.005 (-0.73)
<i>retail</i>					-0.011 (-1.88)	0.002 (0.53)
<i>z_Score</i>						2.726*** (77.22)
<i>year fixed effect</i>	yes	yes	yes	yes	yes	yes
<i>industry fixed effect</i>	yes	yes	yes	yes	yes	yes
<i>Intercept</i>	6.079*** (3.75)	8.555*** (5.21)	8.128*** (4.95)	8.522*** (5.21)	6.412*** (9.18)	-2.963*** (-5.02)
<i>Adjusted R-sq</i>	0.3007	0.3075	0.3080	0.3126	0.3493	0.5645
<i>N</i>	14458	14462	14458	14458	12672	11905

p < .001, "****"; *p* < .01, "***"; *p* < .05, "**"; *p* >= .05, " "

Table-5.2 (Tobin's 'q' as Firm Performance Measure)

This table provides the result of regression on the panel data considering Tobin's 'q' ratio as the dependent variable. Model (1) uses 'pfgrowth' as the independent variable, while considering 'log_ta', 'salesgrowth', 'adv_int', 'rnd_int', 'leverage', and 'log_age' as control variables. Model (2) uses the level 'log_provepx' as the independent variable, while using the same control variables as in model (1). Model (3) uses both 'pfgrowth' and 'log_provepx' as independent variables and has the same controls as in previous models. Model (4) expands model (3) by including ownership dummy variables 'bg', 'psu', and 'fgn'. Model (5) further adds 'insider', 'fii', and 'retail' holdings as control variables. Finally, the model (6) gives the full model by also incorporating z_score as a control variable. Regression coefficients have been given along with significance levels; and t-statistics have been reported in the parenthesis below the coefficients.

	(1)	(2)	(3)	(4)	(5)	(6)
<i>pfgrowth</i>	0.0008 (1.25)		-0.0001 (-0.21)	-0.0002 (-0.28)	-0.0007 (-1.05)	0.00003 (0.06)
<i>log_provepx</i>		0.155*** (9.29)	0.245*** (9.22)	0.246*** (9.09)	0.259*** (9.42)	-0.018 (-0.88)
<i>log_ta</i>	0.218*** (11.32)	0.014 (0.47)	0.013 (0.43)	0.026 (0.86)	-0.187*** (-5.44)	0.166*** (6.14)
<i>salesgrowth</i>	0.010*** (9.75)	0.010*** (10.27)	0.010*** (9.99)	0.010*** (9.99)	0.009*** (8.93)	-0.003*** (-4.09)
<i>adv_int</i>	0.301*** (22.65)	0.288*** (21.63)	0.288*** (21.62)	0.283*** (21.14)	0.254*** (18.54)	0.164*** (15.25)
<i>rnd_int</i>	0.399*** (15.81)	0.374*** (14.76)	0.374*** (14.75)	0.379*** (14.99)	0.373*** (14.64)	0.252*** (12.59)
<i>leverage</i>	-0.031*** (-19.87)	-0.029*** (-18.37)	-0.029*** (-18.35)	-.028*** (-17.56)	-0.024*** (-14.32)	0.019*** (13.77)
<i>log_age</i>	-0.166*** (-3.37)	-0.314*** (-6.14)	-0.316*** (-6.12)	-0.322*** (-6.23)	-0.228*** (-4.28)	-0.065 (-1.60)
<i>bg</i>				-0.068 (-1.07)	-0.032 (-0.49)	0.187*** (3.72)
<i>psu</i>				-0.178 (-1.02)	-0.136 (-0.73)	-0.025 (-0.17)
<i>fgn</i>				0.767*** (5.42)	0.674*** (4.62)	0.611*** (5.46)
<i>insider</i>					0.013*** (5.95)	-0.0007 (-0.42)
<i>fii</i>					0.057*** (12.73)	0.008* (2.50)
<i>retail</i>					-0.011*** (-3.53)	-0.001 (-0.52)
<i>z_Score</i>						1.447*** (77.56)
<i>year fixed effect</i>	yes	yes	yes	yes	yes	yes
<i>industry fixed effect</i>	yes	yes	yes	yes	yes	yes
<i>Intercept</i>	1.066 (1.03)	2.498* (2.39)	2.510* (2.40)	2.394* (2.29)	2.314*** (6.00)	-4.464*** (-14.30)
<i>Adjusted R-sq</i>	0.1486	0.1550	0.1550	0.1571	0.1751	0.4621
<i>N</i>	12885	12887	12885	12885	12496	11905

*p < .001, "****"; p < .01, "***"; p < .05, "**"; p >= .05, " "*

Table-6 (Robustness check using RoE as performance measure)

This table provides the result of regression on the panel data considering RoE ratio as the dependent variable. Model (1) uses 'pfgrowth' as the independent variable, while considering 'log_ta', 'salesgrowth', 'adv_int', 'rnd_int', 'leverage', and 'log_age' as control variables. Model (2) uses the level 'log_provexp' as the independent variable, while using the same control variables as in model (1). Model (3) uses both 'pfgrowth' and 'log_provexp' as independent variables and has the same controls as in previous models. Model (4) expands model (3) by including ownership dummy variables 'bg', 'psu', and 'fgn'. Model (5) further adds 'insider', 'fii', and 'retail' holdings as control variables. Finally, the model (6) gives the full model by also incorporating z_score as a control variable. Regression coefficients have been given along with significance levels; and t-statistics have been reported in the parenthesis below the coefficients.

	(1)	(2)	(3)	(4)	(5)	(6)
<i>pfgrowth</i>	0.019*** (4.36)		0.016*** (3.68)	0.015*** (3.49)	0.009 (1.95)	0.009* (2.45)
<i>log_provexp</i>		0.791*** (4.70)	0.648*** (3.80)	0.872*** (5.03)	0.856*** (4.49)	-0.044 (-0.28)
<i>log_ta</i>	1.013*** (8.12)	0.360 (1.91)	0.471* (2.49)	0.680*** (3.54)	0.720** (3.03)	1.204*** (6.02)
<i>salesgrowth</i>	0.188*** (30.02)	0.195*** (32.38)	0.188*** (30.13)	0.187*** (29.99)	0.198*** (28.54)	0.134*** (22.24)
<i>adv_int</i>	0.342*** (4.01)	0.313*** (3.64)	0.312*** (3.64)	0.296*** (3.44)	0.163 (1.71)	-0.185* (-2.33)
<i>rnd_int</i>	-0.174 (-1.03)	-0.251 (-1.48)	-0.240 (-1.42)	-0.265 (-1.57)	-0.343 (-1.93)	-0.371* (-2.51)
<i>leverage</i>	-0.395*** (-38.61)	-0.389*** (37.53)	-0.390*** (-37.71)	-0.406*** (-38.70)	-0.436*** (-37.93)	-0.161*** (-15.28)
<i>log_age</i>	-0.240 (-0.76)	-0.806* (-2.44)	-0.635 (-1.90)	-0.481 (-1.44)	0.516 (1.40)	0.867* (2.86)
<i>bg</i>				-1.458*** (-3.48)	-1.732*** (-3.82)	0.105 (0.28)
<i>psu</i>				-8.405*** (-7.32)	-12.617*** (-9.81)	-7.123*** (-6.68)
<i>fgn</i>				-5.180*** (-5.49)	-6.103*** (-6.00)	-4.663*** (-5.64)
<i>insider</i>					0.097*** (6.33)	0.023 (1.82)
<i>fii</i>					0.103*** (3.29)	-0.099*** (-3.82)
<i>retail</i>					-0.037 (-1.74)	-0.018 (-1.03)
<i>z_Score</i>						5.931*** (43.05)
<i>year fixed effect</i>	yes	yes	yes	yes	yes	yes
<i>industry fixed effect</i>	yes	yes	yes	yes	yes	yes
<i>Intercept</i>	17.513** (2.94)	22.636*** (3.75)	21.367*** (3.54)	22.460*** (3.73)	7.809** (2.93)	-12.108*** (-5.26)
<i>Adjusted R-sq</i>	0.1876	0.1896	0.1893	0.1932	0.2149	0.3266
<i>N</i>	14458	14462	14458	14458	12672	11905

*p < .001, "****"; p < .01, "***"; p < .05, "**"; p >= .05, " "*

**Table -7.1 (Robustness check using scaled level and difference provident fund expense variables
using ROA as firm performance measure)**

This table provides the result of regression on the panel data considering RoA ratio as the dependent variable. Model (1) uses 'sc_diff_pf' as the independent variable, while considering 'log_ta', 'salesgrowth', 'adv_int', 'rnd_int', 'leverage', and 'log_age' as control variables. Model (2) uses the level 'sc_provexp' as the independent variable, while using the same control variables as in model (1). Model (3) uses both 'sc_diff_pf' and 'sc_provexp' as independent variables and has the same controls as in previous models. Model (4) expands model (3) by including ownership dummy variables 'bg', 'psu', and 'fgn'. Model (5) further adds 'insider', 'fii', and 'retail' holdings as control variables. Finally, the model (6) gives the full model by also incorporating z_score as a control variable. Regression coefficients have been given along with significance levels; and t-statistics have been reported in the parenthesis below the coefficients.

	(1)	(2)	(3)	(4)	(5)	(6)
<i>sc_diff_pf</i>	205.418*** (7.50)		191.831*** (6.43)	177.175*** (5.95)	146.087*** (4.63)	94.882*** (3.74)
<i>sc_provexp</i>		49.764*** (4.02)	15.527 (1.15)	36.525** (2.69)	56.755*** (3.93)	-63.177*** (-5.34)
<i>log_ta</i>	0.456*** (13.42)	0.474*** (13.70)	0.463*** (13.39)	0.640*** (16.54)	0.501*** (10.12)	0.410*** (10.16)
<i>salesgrowth</i>	0.053*** (31.40)	0.055*** (33.47)	0.053*** (31.38)	0.052*** (31.25)	0.055*** (30.57)	0.027*** (17.52)
<i>adv_int</i>	0.212*** (9.13)	0.209*** (8.91)	0.210*** (8.98)	0.201*** (8.63)	0.154*** (6.20)	-0.038 (-1.88)
<i>rnd_int</i>	-0.031 (-0.69)	-0.039 (-0.86)	-0.035 (-0.76)	-0.032 (-0.71)	-0.055 (-1.17)	-0.116** (-3.06)
<i>leverage</i>	-0.175*** (-62.54)	-0.174*** (-61.90)	-0.174*** (-62.07)	-0.179*** (-63.03)	-0.182*** (-60.28)	-0.091*** (-33.51)
<i>log_age</i>	-0.180* (-2.10)	-0.283*** (-3.19)	-0.209* (-2.34)	-0.130 (-1.44)	0.170 (1.78)	0.243** (3.15)
<i>bg</i>				-0.751*** (-6.61)	-0.788*** (-6.67)	-0.098 (-1.03)
<i>psu</i>				-3.102*** (-9.97)	-4.114*** (-12.23)	-3.092*** (-11.37)
<i>fgn</i>				-1.120*** (-4.37)	-1.214*** (-4.55)	-1.089*** (-5.15)
<i>insider</i>					0.040*** (9.85)	0.010** (3.13)
<i>fii</i>					0.079*** (9.57)	-0.005 (-0.81)
<i>retail</i>					-0.010 (-1.78)	0.003 (0.58)
<i>z_Score</i>						2.736*** (77.63)
<i>year fixed effect</i>	yes	yes	yes	yes	yes	yes
<i>industry fixed effect</i>	yes	yes	yes	yes	yes	yes
<i>Intercept</i>	6.113*** (3.77)	6.118*** (3.77)	6.065*** (3.74)	5.824*** (3.61)	3.542*** (5.59)	-2.190*** (-4.13)
<i>Adjusted R-sq</i>	0.3030	0.3017	0.3035	0.3070	0.3457	0.5646
<i>N</i>	14462	14462	14462	14462	12674	11905

p < .001, "***"; *p* < .01, "**"; *p* < .05, "*"; *p* >= .05, " "

Table -7.2 (Robustness check using scaled level and difference provident fund expense variables
using Tobin's 'q' as firm performance measure)

This table provides the result of regression on the panel data considering Tobin's 'q' ratio as the dependent variable. Model (1) uses 'sc_diff_pf' as the independent variable, while considering 'log_ta', 'salesgrowth', 'adv_int', 'rnd_int', 'leverage', and 'log_age' as control variables. Model (2) uses the level 'sc_provexp' as the independent variable, while using the same control variables as in model (1). Model (3) uses both 'sc_diff_pf' and 'sc_provexp' as independent variables and has the same controls as in previous models. Model (4) expands model (3) by including ownership dummy variables 'bg', 'psu', and 'fgn'. Model (5) further adds 'insider', 'fii', and 'retail' holdings as control variables. Finally, the model (6) gives the full model by also incorporating z_score as a control variable. Regression coefficients have been given along with significance levels; and t-statistics have been reported in the parenthesis below the coefficients.

	(1)	(2)	(3)	(4)	(5)	(6)
<i>sc_diff_pf</i>	101.414*** (6.38)		51.180** (2.99)	51.027** (2.98)	40.250* (2.32)	30.528* (2.27)
<i>sc_provexp</i>		69.021*** (9.62)	60.303*** (7.79)	59.989*** (7.65)	64.960*** (8.19)	-6.962 (-1.11)
<i>log_ta</i>	0.221*** (11.48)	0.253*** (12.95)	0.250*** (12.77)	0.259*** (11.84)	0.061* (2.23)	0.149*** (6.96)
<i>salesgrowth</i>	0.009*** (9.03)	0.010*** (10.43)	0.009*** (9.61)	0.009*** (9.62)	0.008*** (8.50)	-0.004*** (-4.53)
<i>adv_int</i>	0.299*** (22.56)	0.287*** (21.61)	0.289*** (21.67)	0.283*** (21.18)	0.254*** (18.53)	0.164*** (15.26)
<i>rnd_int</i>	0.398*** (15.77)	0.382*** (15.16)	0.384*** (15.21)	0.389*** (15.44)	0.383*** (15.08)	0.252*** (12.60)
<i>leverage</i>	-0.031*** (-19.65)	-0.029*** (-18.68)	-0.030*** (-18.72)	-0.029*** (-17.82)	-0.024*** (-14.60)	0.019*** (13.77)
<i>log_age</i>	-0.165*** (-3.37)	-0.294*** (-5.82)	-0.275*** (-5.42)	-0.285*** (-5.57)	-0.190*** (-3.62)	-0.063 (-1.55)
<i>bg</i>				-0.035 (-0.55)	0.002 (0.03)	0.188*** (3.74)
<i>psu</i>				-0.119 (-0.69)	-0.075 (-0.41)	-0.029 (-0.20)
<i>fgn</i>				0.802*** (5.68)	0.710*** (4.88)	0.610*** (5.46)
<i>insider</i>					0.014*** (6.09)	-0.0007 (-0.42)
<i>fii</i>					0.058*** (12.78)	0.009* (2.50)
<i>retail</i>					-0.011*** (-3.41)	-0.001 (-0.46)
<i>z_Score</i>						1.445*** (77.52)
<i>year fixed effect</i>	yes	yes	yes	yes	yes	yes
<i>industry fixed effect</i>	yes	yes	yes	yes	yes	yes
<i>Intercept</i>	1.026 (0.99)	0.831 (0.80)	0.832 (0.80)	0.716 (0.69)	0.383 (1.10)	-4.348*** (-15.50)
<i>Adjusted R-sq</i>	0.1511	0.1545	0.1550	0.1572	0.1749	0.4622
<i>N</i>	12887	12887	12887	12887	12498	11905

*p < .001, "****"; p < .01, "***"; p < .05, "**"; p >= .05, " "*

Table -8.1 (RoA as firm performance measure with robust standard errors)

This table provides the result of regression on the panel data considering 'RoA' as the dependent variable. Standard errors have been adjusted for 58 clusters in 'nic2d'. Model (1) uses 'pfgrowth' as the independent variable, while considering 'log_ta', 'salesgrowth', 'adv_int', 'rnd_int', 'leverage', and 'log_age' as control variables. Model (2) uses the level 'log_provexp' as the independent variable, while using the same control variables as in model (1). Model (3) uses both 'pfgrowth' and 'log_provexp' as independent variables and has the same controls as in previous models. Model (4) expands model (3) by including ownership dummy variables 'bg', 'psu', and 'fgn'. Model (5) further adds 'insider', 'fii', and 'retail' holdings as control variables. Finally, the model (6) gives the full model by also incorporating z_score as a control variable. Regression coefficients have been given along with significance levels; and t-statistics have been reported in the parenthesis below the coefficients.

	(1)	(2)	(3)	(4)	(5)	(6)
<i>pfgrowth</i>	0.007*** (6.26)		0.006*** (4.69)	0.005*** (4.37)	0.004** (2.88)	0.004*** (3.42)
<i>log_provexp</i>		0.386* (2.39)	0.345* (2.08)	0.444* (2.62)	0.418* (2.32)	-0.085 (-0.81)
<i>log_ta</i>	0.444** (3.15)	0.125 (1.01)	0.156 (1.25)	0.260^ (1.97)	0.126 (0.89)	0.513*** (4.05)
<i>salesgrowth</i>	0.052*** (10.42)	0.055*** (10.68)	0.053*** (10.10)	0.052*** (10.32)	0.055*** (11.02)	0.027*** (8.37)
<i>adv_int</i>	0.214^ (1.80)	0.199 (1.67)	0.198 (1.65)	0.189 (1.64)	0.145 (1.27)	-0.045 (-0.72)
<i>rnd_int</i>	-0.029 (-0.15)	-0.067 (-0.33)	-0.063 (-0.32)	-0.066 (-0.33)	-0.082 (-0.48)	-0.120 (-1.15)
<i>leverage</i>	-0.175*** (-16.66)	-0.172*** (-17.30)	-0.173*** (-17.51)	-0.178*** (-17.66)	-0.180*** (-18.97)	-0.091*** (-6.79)
<i>log_age</i>	-0.147 (-0.67)	-0.419^ (-1.92)	-0.357 (-1.59)	-0.284 (-1.19)	0.050 (0.19)	0.201 (0.84)
<i>bg</i>				-0.863*** (-3.43)	-0.88** (-3.15)	-0.123 (-0.62)
<i>psu</i>				-3.405*** (-3.83)	-4.338*** (-4.02)	-3.167** (-2.70)
<i>fgn</i>				-1.262* (-2.05)	-1.326^ (-1.83)	-1.114* (-2.04)
<i>insider</i>					0.039** (3.26)	0.010^ (1.71)
<i>fii</i>					0.079*** (4.88)	-0.005 (-0.36)
<i>retail</i>					-0.011 (-1.14)	0.002 (0.32)
<i>z_Score</i>						2.727*** (8.09)
<i>year fixed effect</i>	yes	yes	yes	yes	yes	yes
<i>industry fixed effect</i>	yes	yes	yes	yes	yes	yes
<i>Intercept</i>	6.078*** (3.92)	8.555*** (5.20)	8.128*** (4.85)	8.522*** (5.12)	6.413*** (4.26)	-2.962* (-2.25)
<i>Adjusted R-sq</i>	0.3007	0.3075	0.3080	0.3126	0.3493	0.5645
<i>N</i>	14458	14462	14458	14458	12672	11905

p < .001, "***"; *p* < .01, "**"; *p* < .05, "*"; *p* < .10, "^", *p* >= .10, ""

Table -8.2 (Tobin's 'q' as firm performance measure with robust standard errors)

This table provides the result of regression on the panel data considering Tobin's 'q' as the dependent variable. Standard errors have been adjusted for 58 clusters in 'nic2d'. Model (1) uses 'pfgrowth' as the independent variable, while considering 'log_ta', 'salesgrowth', 'adv_int', 'rnd_int', 'leverage', and 'log_age' as control variables. Model (2) uses the level 'log_provexp' as the independent variable, while using the same control variables as in model (1). Model (3) uses both 'pfgrowth' and 'log_provexp' as independent variables and has the same controls as in previous models. Model (4) expands model (3) by including ownership dummy variables 'bg', 'psu', and 'fgn'. Model (5) further adds 'insider', 'fii', and 'retail' holdings as control variables. Finally, the model (6) gives the full model by also incorporating z_score as a control variable. Regression coefficients have been given along with significance levels; and t-statistics have been reported in the parenthesis below the coefficients.

	(1)	(2)	(3)	(4)	(5)	(6)
<i>pfgrowth</i>	0.0008 [^] (1.68)		-0.0001 (-0.38)	-0.0002 (-0.53)	-0.0007* (-2.07)	0.00002 (0.07)
<i>log_provexp</i>		0.243** (2.83)	0.245** (2.85)	0.247** (3.18)	0.259*** (3.51)	-0.019 (-0.32)
<i>log_ta</i>	0.218** (2.87)	0.013 (0.11)	0.013 (0.10)	0.026 (0.19)	-0.187 (-1.21)	0.166 [^] (1.74)
<i>salesgrowth</i>	0.010*** (9.54)	0.010*** (9.46)	0.010*** (9.72)	0.010*** (9.98)	0.009*** (8.86)	-0.003* (-2.51)
<i>adv_int</i>	0.301 [^] (1.85)	0.288 [^] (1.75)	0.288 [^] (1.75)	0.283 [^] (1.76)	0.254 (1.64)	0.164 [^] (1.68)
<i>rnd_int</i>	0.399 (1.15)	0.374 (1.12)	0.374 (1.12)	0.379 (1.14)	0.373 (1.10)	0.252 (1.28)
<i>leverage</i>	-0.031*** (-7.86)	-0.029*** (-7.58)	-0.029*** (-7.57)	-0.028*** (-7.41)	-0.024*** (-6.43)	0.020*** (5.07)
<i>log_age</i>	-0.166 (-1.58)	-0.314* (-2.09)	-0.315* (-2.09)	-0.322* (-2.07)	-0.227 (-1.54)	-0.066 (-0.83)
<i>bg</i>				-0.068 (-0.50)	-0.032 (-0.23)	0.188* (2.10)
<i>psu</i>				-0.178 (-0.38)	-0.136 (-0.30)	-0.025 (-0.07)
<i>fgn</i>				0.767* (2.04)	0.674 [^] (1.70)	0.611* (2.07)
<i>insider</i>					0.013** (2.87)	-0.0007 (-0.30)
<i>fii</i>					0.057*** (5.05)	0.009 (1.60)
<i>retail</i>					-0.011 [^] (-1.83)	-0.001 (-0.33)
<i>z_Score</i>						1.447*** (9.84)
<i>year fixed effect</i>	yes	yes	yes	yes	yes	yes
<i>industry fixed effect</i>	yes	yes	yes	yes	yes	yes
<i>Intercept</i>	1.066 [^] (1.98)	2.498** (2.84)	2.509** (2.85)	2.395** (2.67)	2.314 (1.55)	-4.464*** (-3.84)
<i>Adjusted R-sq</i>	0.1486	0.1550	0.1550	0.1571	0.1751	0.4621
<i>N</i>	12885	12887	12885	12885	12496	11905

*p < .001, "****"; p < .01, "***"; p < .05, "**"; p < .10, " ^ ", p >= .10, " "*

Table -9 (Hausman Test)

This table gives the result of Hausman test carried out to compare suitability of fixed effect vis-à-vis random effect model. The null being tested is that difference between coefficients is not systematic; hence a significant test statistic would mean that assumptions regarding fixed and random effect models are systematically different and a fixed effect model would be better suited. The chi square statistics are given in rows, while corresponding probability values are reported in parentheses. Number of variables involved in the model is also reported.

	Model (3)	Model (5)	Model (6)
RoA	9.52 (0.2177)	69.47 (0.0000)	33.55 (0.0024)
Tobin's 'q'	58.63 (0.0000)	107.85 (0.0000)	335.53 (0.0000)
RoE	28.82 (0.0002)	146.57 (0.0000)	365.58 (0.0000)
Number of variables	7	13	14

Table -10.1 (2 Step Least Square Regression)

This table reports the results of two-stage least square regressions using instrumental variables. 'pfgrowth' and 'log_provexp' have been predicted using the instrumental variables 'empexpgrowth' and 'log_netempexp' in the first stage; the dependent variables 'RoA', Tobin's 'q', and RoE have then been regressed using these predicted values of 'pfgrowth' and 'log_provexp' in second stage.

	RoA	Tobin's 'q'	RoE
<i>pfgrowth</i>	0.049*** (9.87)	0.007*** (3.62)	0.151*** (7.85)
<i>log_provexp</i>	0.808*** (12.30)	0.369*** (14.10)	1.401*** (6.07)
<i>log_ta</i>	-0.544*** (-8.31)	-0.239*** (-6.08)	-1.067*** (-4.04)
<i>salesgrowth</i>	0.047*** (14.32)	0.006*** (4.60)	0.180*** (12.76)
<i>adv_int</i>	0.062* (1.97)	0.223*** (8.28)	-0.036 (-0.36)
<i>rnd_int</i>	0.039 (0.65)	0.305** (2.83)	-0.151 (-0.74)
<i>leverage</i>	-0.159*** (-48.92)	-0.025*** (-19.45)	-0.360*** (-22.13)
<i>log_age</i>	0.317** (2.83)	-0.168^ (-1.94)	2.002*** (5.10)
<i>bg</i>	-0.274* (-2.08)	-0.055 (-0.79)	0.600 (1.25)
<i>psu</i>	-2.058*** (-6.14)	-0.293^ (-1.65)	-5.663*** (-4.37)
<i>fgn</i>	-0.490 (-1.46)	0.572*** (3.23)	-3.789*** (-3.42)
<i>insider</i>	0.028*** (6.72)	0.015*** (9.24)	0.071*** (4.51)
<i>fii</i>	0.080*** (9.00)	0.061*** (14.53)	0.115*** (3.52)
<i>retail</i>	-0.008 (-1.26)	-0.008*** (-3.30)	-0.031 (-1.28)
<i>N</i>	12672	12496	12672

*p < .001, "****"; p < .01, "***"; p < .05, "**"; p < .10, " ^ ", p >= .10, " "*

Table -10.2 (Tests for endogeneity)

This table reports the Durbin and Wu-Hausman chi-square test statistic to check the null hypothesis that variables are exogenous. Test statistics have been noted for all the three performance measures considered, viz., RoA, Tobin's 'q', and RoE. Corresponding probability values have been given in parenthesis below the respective statistic.

Dependent Variable	Durbin Test Statistic	Wu Hausman Test Statistic
RoA	56.2144 (0.0000)	28.2181 (0.0000)
Tobin's 'q'	135.384 (0.0000)	68.1742 (0.0000)
RoE	70.5076 (0.0000)	36.2258 (0.0000)

Table -10.3 (Strength of Instruments used)

This table presents the robust F-test to check the strength of the instrumental variables used. The null hypothesis is that the instruments are weak. Corresponding probability values have been given in the parenthesis.

	RoA/RoE	Tobin's 'q'
pfgrowth	421.573 (0.0000)	411.37 (0.0000)
log_provexp	8981.19 (0.0000)	8899.62 (0.0000)
Degrees of Freedom	(2, 12657)	(2, 12481)